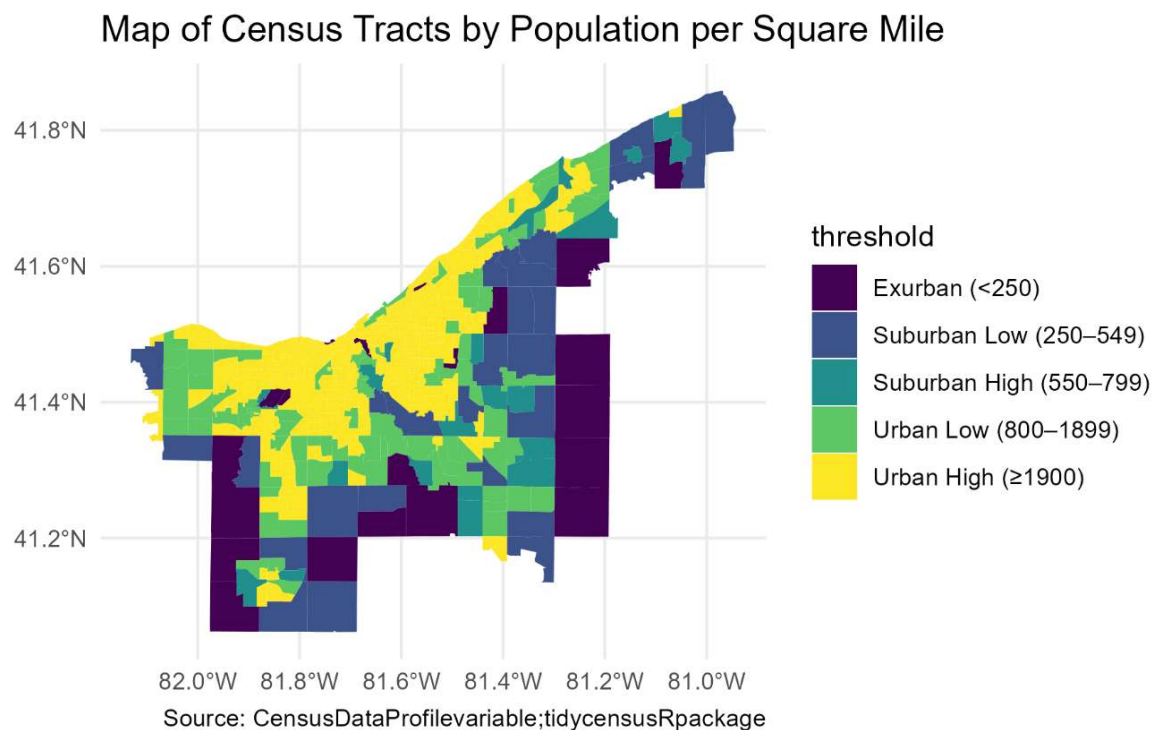


Replication Study

I chose to examine the Greater Cleveland metropolitan area. The metropolitan area has a population of about 2.2 million people, as of the 2020 census. Cleveland, the largest city within the metro at about 370,000 people, anchors the region on the south shore of Lake Erie, while the rest of the metro radiates outward. As a whole, the region has a median income of about \$46,000 with about 15% of people living below the poverty line. Demographically, the area is approximately 71% White, 20% Black, 5% Hispanic/Latino, 2% broadly Asian, with the remaining comprised of two or more races, Native Americans and Alaskan Natives. Gender differences are almost evenly split, with about 52% of the population identifying as female and 48% as male.

Figure 1 shows the spatial distribution of population density follows a pattern radiating out from Cleveland. Generally, the thresholds move from most dense to least dense the farther away they are from the central city. This creates a concentric ring effect where lower-density rings surround higher ones, a pattern seen throughout the country resulting from suburbanization. That said, with the thresholds structured as they are, the graph loses deeper nuance that might show subtler differences between the thresholds.

Figure 1 Map of Population Thresholds



Figures 2 and 3 compare the populations between the suburban and urban thresholds by male and female genders.

Figure 2 shows the suburban distribution having relatively equal younger (under 24) and older (over 65) people. However, there is a trend, starting at about 40 years old of increasing people until 65, when it drops off steeply. This could indicate a relocation out of the suburbs, fitting existing cultural norms of retirement in warm weather. Figure 3, shows the urban distribution having roughly equal populations, with bulges in the 25 to 29 range and 55 to 64 range. After 64, there is a marked drop off in people 65 years old and over.

Figure 2: Suburban Population Pyramid

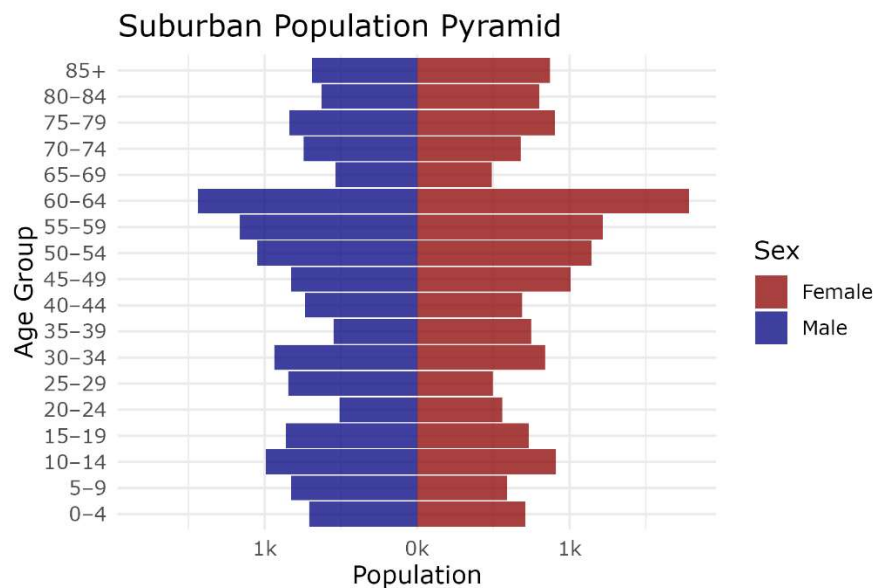


Figure 3: Urban Population Pyramid

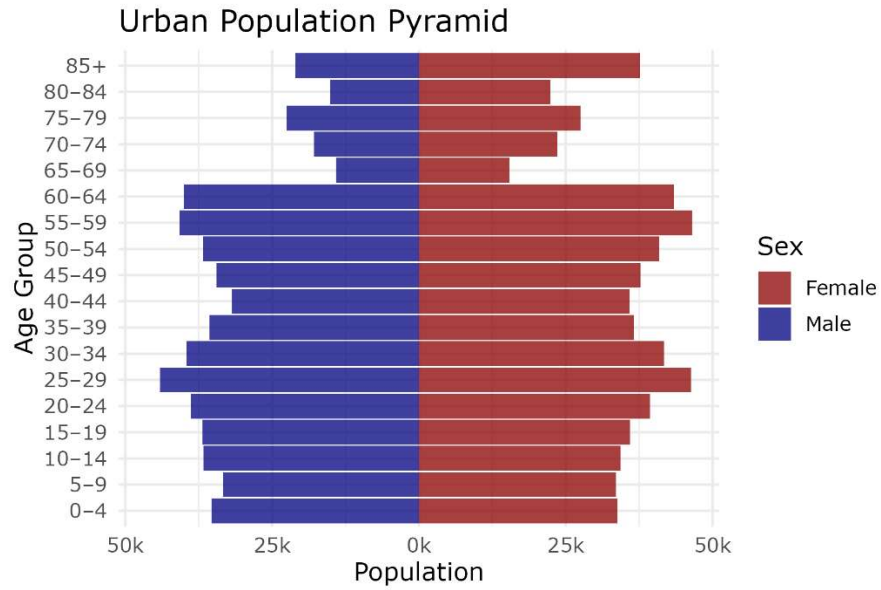
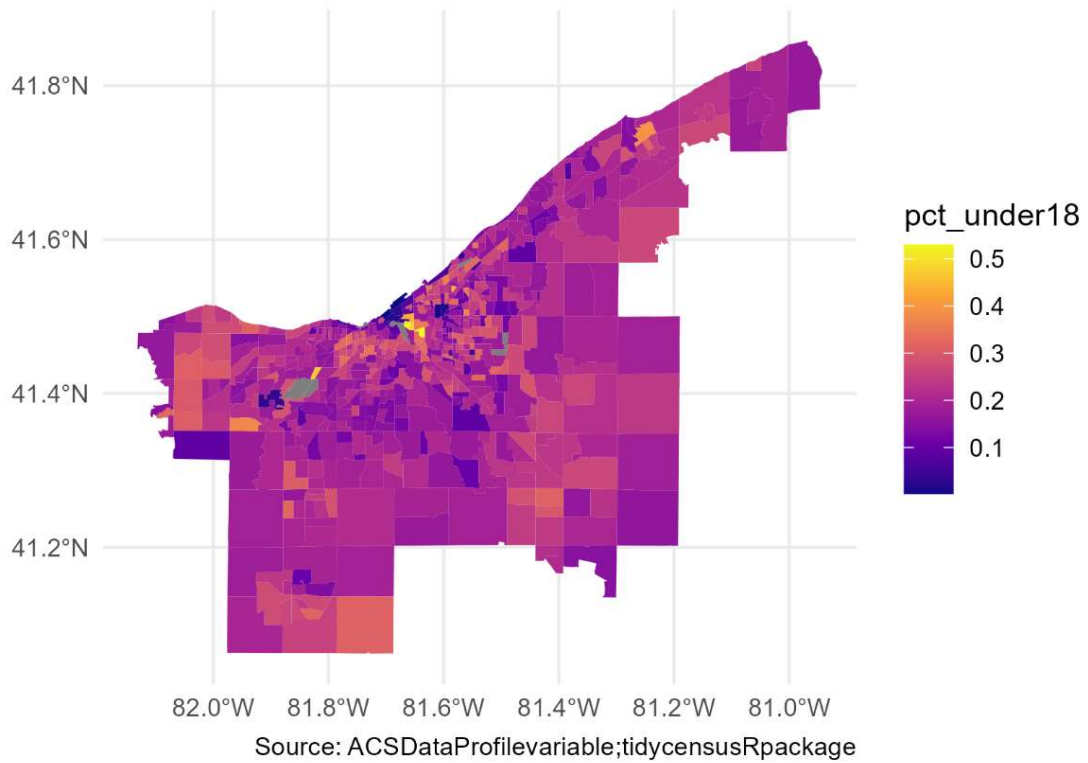


Figure 4: Map of Population Under 18 shows the proportion of Census tracts population that is under 18, across the metropolitan area. This indicates there are some rough spatial patterns, with a ring of higher percent around the core city, followed by a ring of lower percent, and then another higher one.

Figure 4: Map of Population Under 18

Map of Census Tracts by Percent of Population under 18



To test the relationship between tract density and proportion of children, I ran three regressions. Figures 5, 6, and 7 show regressions of the relationship between a tract's density and its percent of the population under 18 years old.

Figure 5 depicts Model 1, a simple linear regression that indicates a slight negative relationship with extreme variation in the percent of under 18-year-olds in tracts of comparable density. Figure 6 provides a regression with a log of the density variable to build Model 2 and indicates a slight positive relationship, again with extreme amounts of variation. Finally, Figure 7 constructed Model 3 by separating the densities by thresholds and shows slight positive relationships for the non-urban spaces and negative ones for the urban ones.

Figure 5: Density and Population under 18

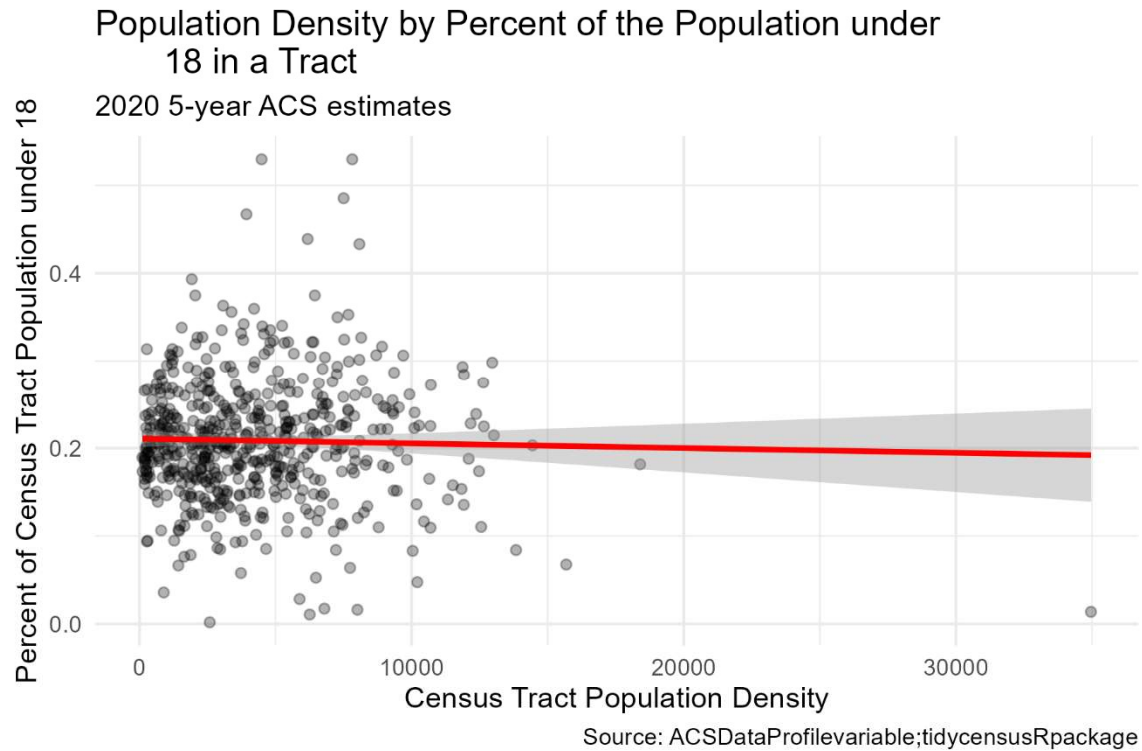


Figure 6: Log Density and Population Under 18

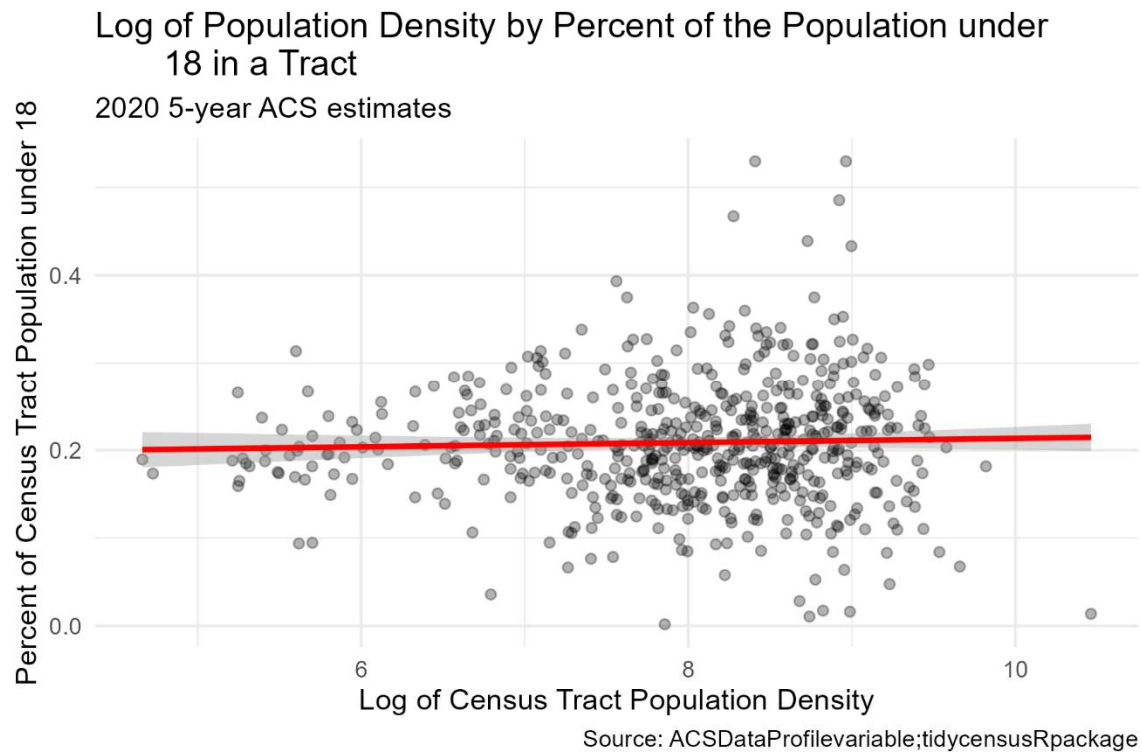


Figure 7: Log Density and Population Under 18 by Threshold

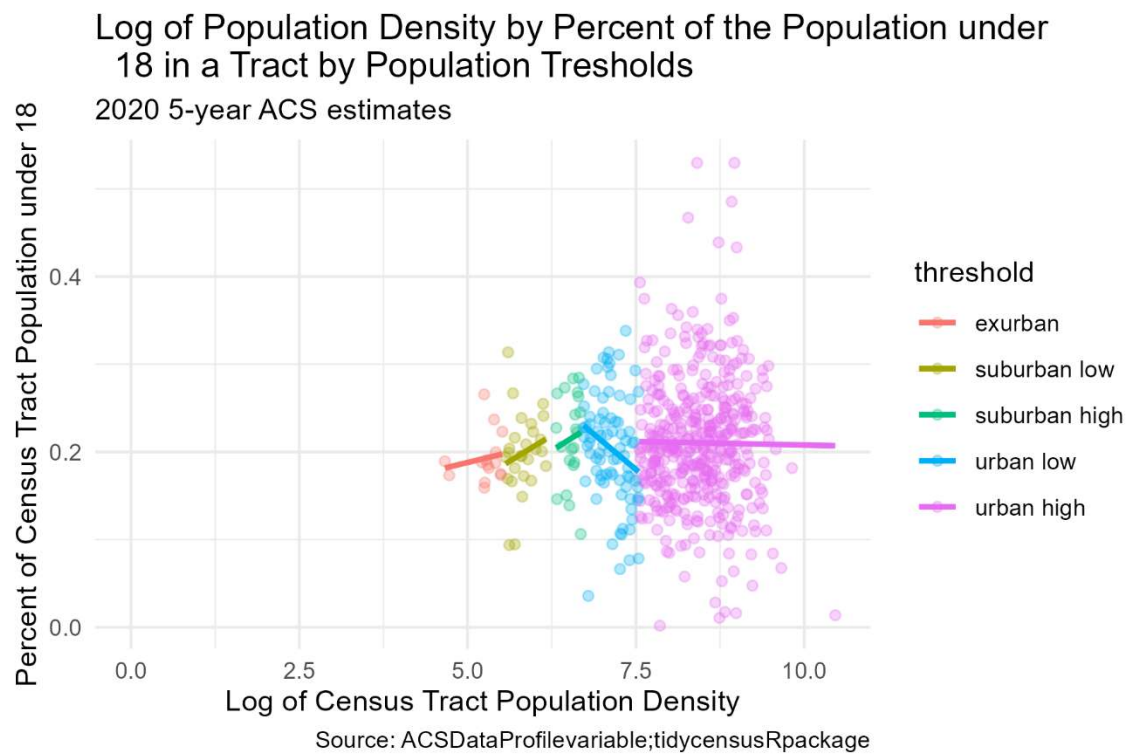


Table 1 provides tests of significance for the three models. Although there appears to be some level of association between the variables used in Figure 5, 6, and 7, the model summary table indicates that none were significant at any level. This is likely due to the large amount of variation of the percent of children within each tract at similar densities. What we can learn from this is that density of the urban environment is not a strong predictor for the population breakdown with regard to the number of children.

Table 1: Model Summaries

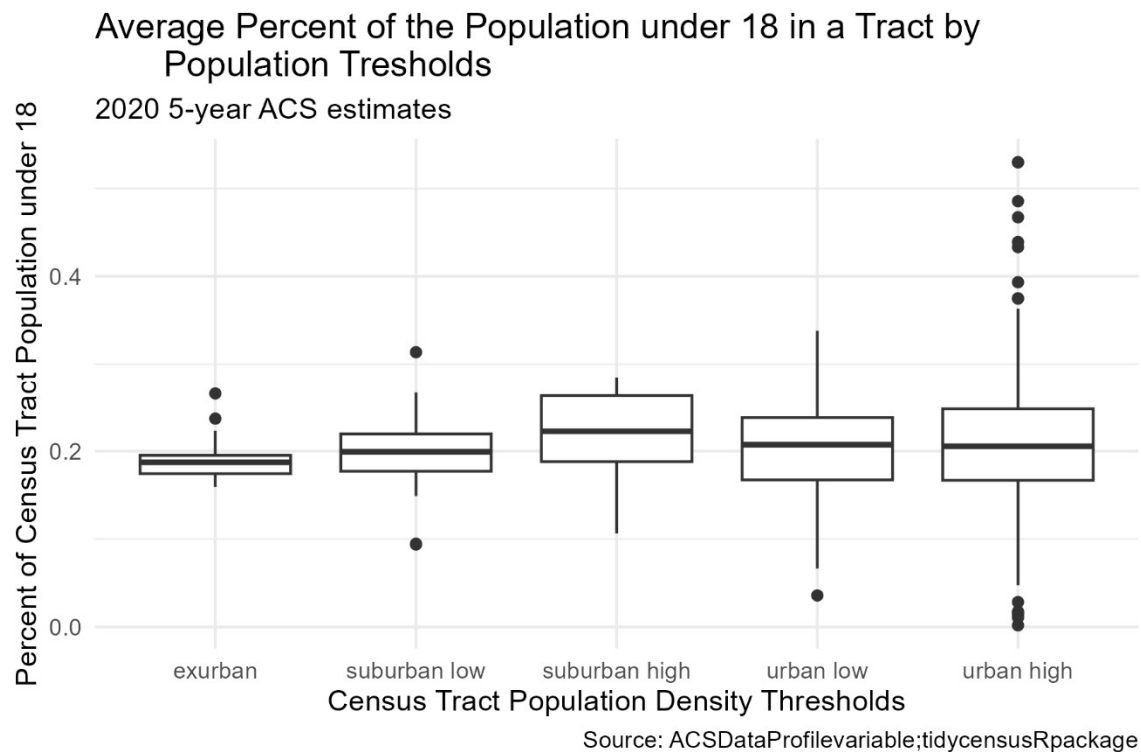
	model1	model2	model3
Intercept	0.21***	0.19***	0.10
	(0.00)	(0.02)	(0.39)
Tract Density	-0.00		
	(0.00)		

	model1	model2	model3
Log Density		0.00	0.02
		(0.00)	(0.07)
Suburban (Low)			-0.17
			(0.57)
Suburban (High)			-0.20
			(0.96)
Urban (Low)			0.57
			(0.45)
Urban (High)			0.13
			(0.39)
Log Density × Suburban (Low)			0.03
			(0.10)
Log Density × Suburban (High)			0.03
			(0.15)
Log Density × Urban (Low)			-0.08
			(0.08)
Log Density × Urban (High)			-0.02
			(0.07)
Num.Obs.	574	574	574

	model1	model2	model3
R2	0.001	0.001	0.012
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

Figure 6 compares the thresholds against the average density and presents the a similar conclusion to Table 1. There is significant variation in the proportion of a tracts population that is under 18. Although this is particularly true in the urban high category, each category experiences high variation.

Table 2: Average Population under 18 and Threshold



- A link to your GitHub Repository where the code is located.